Aneri Raval

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Summary

Engineer with a research orientation toward **persistent memory for AI agents**: how language-model agents can store, retrieve and reason over knowledge across sessions and long-horizon tasks. Pairs an evidence-first, statistical mindset, from satellite-data machine learning at **ISRO** and hypothesis-driven analysis, with two years of building production AI systems in an enterprise setting. Writes and speaks on AI, and is seeking research-focused roles and graduate study in agent memory and applied machine learning.

Education

Aditya Silver Oak Institute of Technology, Silver Oak University Ahmedabad, India
Bachelor of Engineering in Computer Engineering | CGPA: 9.94/10.0 | University Topper *2020 – 2024*

Technical Skills

- Languages: Java, Python, SQL, C++
- Backend: Spring Boot, FastAPI, REST APIs, Microservices, API Design
- Databases & Messaging: MongoDB, SQL Server, Kafka
- Cloud & DevOps: Microsoft Azure, Docker, GitHub Actions, Dynatrace, SonarQube, Semgrep, Power BI
- AI / ML: LangChain, RAG, FAISS, Scikit-Learn, TensorFlow, Keras, CNN, OpenCV, Computer Vision, Generative AI, AI Agents
- Data & Research: Statistical Analysis, Hypothesis Testing, NumPy, Pandas, Satellite Data Analysis

Experience

Tata Consultancy Services Ahmedabad
Backend Engineer (Client: Marks & Spencer, UK) *Nov 2024 – Present*

- Built and demoed an AI-driven **Spec-Driven Development agent** that generates end-to-end API implementations from specification files per team coding standards, cutting boilerplate development time and freeing engineering focus for system design and security; showcased live to client stakeholders.
- Developed a **GitHub agentic workflow** that automatically triages and remediates SonarQube maintainability issues and Snyk vulnerabilities, replacing manual portal-based review and fixes.
- Engineered a **Notification Framework** (Java, Spring Boot, Kafka, MongoDB) automating multi-channel communication workflows across enterprise retail systems.
- Built and optimized **RESTful microservices** and MongoDB data-access layers handling high-volume retail transactions; integrated KTSL Helix APIs for automated incident management.
- Delivered **Power BI** dashboard integrations for real-time KPI visibility; contributed CI/CD improvements via GitHub Actions and production monitoring via Dynatrace.
- Mentored **100+ associates** on AI/ML practices as part of internal enablement initiatives.

Space Applications Centre (SAC), Indian Space Research Organisation (ISRO) Ahmedabad
Research Intern | GitHub *Jun 2023 – Jul 2024*

- Thesis: *Development of Machine Learning Methodologies to Detect Deep Convective Cloud (DCC) Systems in ScaRaB using SAPHIR on board Megha-Tropiques.*
- Built a **Random Forest** classification model for DCC detection on ScaRaB satellite sensor data, evaluating it against alternative classical ML approaches (SVM, XGBoost) and achieving a **20% accuracy improvement** over baseline, validated through precision-recall analysis on held-out test sets.
- Processed and feature-engineered **16+ GB** of HDF satellite datasets from MOSDAC using NumPy and Pandas; experimented with model architectures for weather and climate prediction.
- Co-authored and presented: A. Raval, K. Patel, M. V. Shukla, *Development of AI/ML Methodology to Detect DCC in SCARAB using SAPHIR on board Megha-Tropiques*, National Symposium on Tropical Meteorology (TROPMET 2023), Indian Meteorological Society, Nov 2023.

BISAG-N (Bhaskaracharya National Institute for Space Applications and Geo-informatics) Gandhinagar
AI/ML Intern | GitHub *Jun 2022 – Jul 2022*

- Built a real-time **Facial Emotion Recognition** system using a CNN (TensorFlow/Keras) and OpenCV, classifying **7 emotion categories** from live webcam input with optimized low-latency predictions.

Key Projects

Diabetes Hypothesis Testing | GitHub

- Designed a **hypothesis-driven** analysis (Python, Jupyter) of diabetes-related factors, defining testable research questions before any modelling.
- Applied group comparisons, **equivalence tests** and adjusted models to control for confounders, and validated predictive performance with **cross-validation**.

Legacy-to-SAP S/4HANA Migration Assistant | TCS AI Fridays Regional Finale – Runner-Up

- Built an AI-assisted data migration solution that helps move legacy-system data into **SAP S/4HANA**, targeting the field-mapping step that is usually done manually.
- Used **LangChain and RAG** for contextual field mapping, combined with a **Random Forest classifier** refined with RLHF-based feedback to improve mapping suggestions over time.
- Deployed the backend on **GCP** with a React (Vite) frontend and a Supabase database; built as a hackathon solution (no public repository).

PaperInsight – AI Research Analyser | GitHub

- Built a **RAG** research assistant (LangChain) that ingests academic PDFs and answers context-aware questions grounded in the paper's own content, instead of relying on a general chatbot.
- Used FAISS for **sub-second vector retrieval** of relevant passages, with Gemini AI generating answers and a Streamlit interface for semantic search across papers.

ML-Integrated Backend Microservices | GitHub

- Architected polyglot microservices (**Spring Boot + FastAPI**) containerized with Docker, where a Java backend consumes a Python ML service through a REST API.
- Served a Scikit-Learn prediction model behind the API, kept model serving and business logic independently deployable, and automated **CI/CD** with GitHub Actions; documented in a Medium article.

Content-Based Movie Recommendation System | GitHub

- Built a **content-based recommender** that suggests similar movies from metadata alone (genres, keywords, cast, director and taglines), so new titles can be recommended without any user history.
- Vectorized the combined metadata with **TF-IDF** and ranked movies by **cosine similarity**.

Honors and Awards

- **TCS Beyond Delivery** – Presented the **Spec-Driven Development** initiative to the CEOs and CXOs of TCS and Marks & Spencer; recognized for it during quarterly planning.
- **Special Initiative Award** – TCS Shared Service Group (Aug 2026), for mentoring at the TCS AI Friday Season 2 Regional Finale.
- **2nd Runner-up** – TCS AI Fridays Regional Finale, Season 1 (Dec 2025).
- **Outstanding Alumni Award**, IT & Technology Innovations – Silver Oak University (Mar 2025).
- **University Topper** – Dewang Mehta IT Awards (Apr 2024), with a 9.94 CGPA.

Leadership & Volunteer Experience

Mentor – TCS AI Fridays *2026*

- Mentored next season's participants after a Regional Runner-Up finish; selected as mentor for the following finale.

Speaker – GDG Ahmedabad International Women's Day *2026*

- Presented on *How to Break the Backend Barrier* and *How Backend Developers Can Transition into AI Roles.*

Technical Writer – Medium *medium.com/@ahraval11*

- Write on backend engineering, ML integration and AI: Spec-Driven API development with AI agents, prompt frameworks, JWT authentication and RBAC and JUnit testing in Spring Boot, and CI/CD for ML microservices.

Team Lead – Smart India Hackathon 2022 *2022*

- Led a student team to the Smart India Hackathon 2022 finals.

Vice President – E-Cell IGNITE

- Led entrepreneurship initiatives for a 200+ student community.

Certifications

- Claude Certified Associate – Foundations (CCAO-F)
- Claude Certified Developer – Foundations (CCDV-F)
- GitHub Certified Agentic AI Developer (GH-600)
- GitHub Actions (GH-200)
- Microsoft Certified: Azure Developer Associate (AZ-204)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Hands-on ML-Driven Business Analytics – Indian Institute of Science (IISc), Dec 2024